SECTION 15110-A
MANUAL VALVES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section includes valves and cocks required in all plumbing, hydronic, and steam systems.

B. Specialty valves described in other Division 15 Sections.

1.3 SHOP DRAWINGS

A. Submit detailed shop drawings in accordance with Division 1. Clearly indicate make, model, location, type, size, and pressure rating.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

A. Valves:
   1. According to the recommended manufacturer table.

B. Provide valves of same manufacturer for each valve type. No more than 3 valve manufacturers shall be used for the Project.

C. Provide valves with manufacturer's name and pressure rating clearly marked on outside of body.

D. Packing and gasket materials shall be nonasbestos.

E. Refer to piping sections for specific valve model numbers.

2.2 VALVE CONNECTIONS

A. Provide valves compatible with adjoining piping pipe joints. Use pipe size valves.

B. Thread pipe sizes 2 inches and smaller.

C. Flange pipe sizes 2-1/2 inches and larger.

D. Solder or screw-to-solder adaptors for copper tubing.

2.3 GATE VALVES

A. Bronze Body: Union bonnet, rising stem, inside screw, bronze solid wedge, solder or screwed ends, bronze packing gland, Teflon®-impregnated packing and malleable handwheel.

B. Iron Body: Bolted bonnet, bronze trim, rising stem, OS&Y, cast iron solid wedge, flanged ends, Class 125, Teflon®-impregnated packing and 2-piece packing gland.

2.4 GLOBE OR ANGLE VALVES

A. Bronze Body: Union bonnet, rising stem, inside screw, renewable disc, solder or screwed ends, Class 125, copper silicon alloy stem, brass packing gland, Teflon®-impregnated packing and malleable handwheel.

B. Iron Body: Bolted bonnet, bronze trim, rising stem, OS&Y, renewable seat and disc, flanged ends, Class 125, Teflon®-impregnated packing and 2 piece packing gland.

2.5 BOILER AUTOMATIC STOP-CHECK VALVES

A. Iron body, bronze trim, rising stem, OS&Y, external dampening control, Teflon®-based sealing material, flanged ends.

2.6 CHECK VALVES - VERTICAL INSTALLATION ONLY

A. Bronze Body 2 inches and Smaller: Screwed bonnet, renewable swing disc, solder or screwed ends.

B. Iron Body 2-1/2 inches and Larger: Bolted bonnet, bronze trim, swing disc, renewable seat and disc, flanged ends.

2.7 CHECK VALVES - VERTICAL OR HORIZONTAL

A. Bronze Body 2 inches and Smaller: Stainless steel and bronze trim, spring-loaded silent check, renewable Buna-N disc and Teflon® seat ring, screwed ends, vertical pattern.

B. Iron Body 2-1/2 inches and Larger: Stainless steel and bronze trim, spring-loaded silent check, renewable Buna-N seat and bronze disc, flanged ends, vertical pattern, Class 125/250 iron body.

C. Non-slam, globe type, Iron Body 2-1/2 inches and Large: Bronze trim and seats, ANSI Class 250 end connections, integral trim construction.

2.8 PLUG COCKS

A. Semisteel body and plug, rectangular port, Teflon® surfaces, pressure-lubricated type, lubrication port with check valve, screwed or flanged ends.

2.9 GAS COCKS

A. Iron body, bronze plug and washer, square head, lever handle, screwed ends.

B. Iron body and plug, square head, lever handle, flanged ends.

2.10 BUTTERFLY VALVES

A. Iron body, aluminum/bronze disc, replaceable EPDM, Viton, Buna-N, cartridge type liner with rigid phenolic backing, 200 psi bubble-tight shut-off, full lug pattern, tapped lugs, 410 SS stems, and extended neck. Semilug pattern valves are not acceptable.

2.11 BALL VALVES

A. Bronze body, ASTM B62 material, 3 piece construction, chrome plated ball, threaded packing gland, reinforced Teflon® seats, full port, plated steel handle, solder or screwed ends.

2.12 DRAIN VALVES

A. Refer to Section 15440.

2.13 SPECIAL SYSTEM VALVES

A. Refer to appropriate special system Specification Sections.

2.14 PRESSURE RATING

A. Use valves suitable for 125 psi WSP minimum or 150 percent of system operating pressure, whichever is greater.

B. For fire protection use valves suitable for 175 psi.

2.15 VALVE OPERATORS

A. Provide suitable handwheels for gate, globe, angle, and drain valves and for inside hose bibbs.

B. Provide 1 plug cock wrench for every 10 plug cocks sized 2 inches and smaller, minimum of 1. Provide a wrench with set screw for each plug cock sized 2-1/2 inches and larger.

C. For butterfly valves provide gear operators for sizes 8 inches and larger. For smaller sizes provide latch-lock handle with toothed plate for shutoff service, and infinitely variable handle with lock nut and memory stop for throttling service.

D. Valves 6 inches and larger located more than 7'-0" above floor in equipment room areas shall be provided with chain operated sheaves. Extend chains to approximately 5'-0" above floor and hook to clips arranged to clear walking aisles.

E. Provide cast iron or cast aluminum hand wheels. Gear box covers shall be steel or aluminum. Plastic materials in these locations will not be acceptable.

F. Provide valves installed in insulated piping systems with extended stems to assure full clearance of insulation during operation thereof.

PART 3 -- EXECUTION

3.1 INSTALLATION

A. Provide valves with stems upright or horizontal, not inverted.

B. Provide ball or butterfly valves for shutoff and isolating service, and to isolate equipment, part of systems, and vertical risers, except in steam and condensate return piping systems where a gate valve shall be provided.

C. Provide ball or butterfly valves for throttling service, and in control device or water meter bypasses.

D. Provide vertical check valves where applicable in discharge of chilled water, condenser water, heating water, and domestic water pumps.

E. Provide horizontal check valves where vertical check valves are not compatible with piping arrangements.

F. Do not use horizontal check valves in vertical applications.

G. Provide gas cocks for gas service.

H. Provide plug cocks in water systems for throttling service. Use nonlubricating plug cocks only when shutoff or isolating valves are also provided.

I. Provide drain valves at main shutoff valves and low points of piping and apparatus.

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